

System Design for High-Capacity Unrepeated Optical Transmission

João C. S. S. Januário, Andrea Chiuchiarelli, Sandro M. Rossi, José Hélio C. Junior, Jacklyn D. Reis, Cristiano Mornatta, Alessandro Festa, Aldário C. Bordonalli

Abstract—By basing the figure of merit on the joint evaluation of signal BER and OSNR, the design of single-carrier 400G unrepeated optical systems using remote optically-pumped amplifiers (ROPAs) with dedicated delivery fiber is discussed and analyzed in this work. Simulation results show very good agreement with experimental measurements, proving the effectiveness of the proposed approach in critical transmission situations.

Index Terms—Unrepeated coherent transmission, distributed Raman amplifier (DRA), remote optically-pumped amplifier (ROPA), optical system design.

I. INTRODUCTION

In the global telecommunications scenario, great interest has recently been devoted to connecting areas in which the difficulties imposed by climatic and environmental limitations are not straightforward to overcome. Due to the strong deployment and maintenance constraints associated with the use of in-line active components for challenging submarine and terrestrial applications, unrepeated optical transmission using coherent detection appeared as a breakthrough technology to promote high-speed connectivity in remote and hard-to-reach areas [1]. In this context, system design is crucial and to achieve optimal performance it is important to jointly optimize optical (amplifiers and fibers) and electronic subsystems, such as those employing digital signal processing (DSP).

Several research works on unrepeated transmission of coherent optical signals with 100G and 200G per channel bit rates have been reported over the past few years [2]–[6]. In [2], the authors demonstrated the first high capacity (3.84 Tb/s) unrepeated transmission of 224 Gb/s polarization-division multiplexed (PDM)-16QAM channels, reaching 240 km with 80 50-GHz spaced WDM channels. In [7], an average Q-factor improvement of 2.5 dB using digital backpropagation (DBP) to compensate nonlinear impairments was demonstrated. The application of digital nonlinear compensation in conjunction with first-order backward Raman amplification allowed the transmission of 40×256 Gb/s PDM-16QAM channels (8 Tb/s total capacity) over a 304-km hybrid fiber link.

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Manuscript received X, 2018; revised X, 2018.

In [4], first-order co-propagating Raman amplification together with a backward pumped ROPA stage at the receiver end was used. The transmission of 40×200 Gb/s PDM-16QAM channels (8 Tb/s total capacity) using low-loss fiber allowed a total system reach of 363 km. In [5], the use of a high power booster erbium doped fiber amplifier (EDFA) and third-order Raman amplification enabled the transmissions of 80×200G channels over 321 km with no ROPA and of 16×200G channels over 363 km with the addition of a ROPA to the system setup. The spectral efficiency (SE) achieved in the latter case was 5.33 b/s/Hz (37.5-GHz channel spacing). Those results were then surpassed by [6], where real-time unrepeated transmission of 120×200 Gb/s PDM-16QAM channels (24 Tb/s total capacity) was demonstrated for a system reach of 349 km. A 30-dBm high power booster EDFA in conjunction with ROPA and third-order Raman amplification were used to attain those results. Moreover, a 33.33-GHz (SE = 6 b/s/Hz) channel spacing yielded a record capacity-reach product of 8.38 Pb/s/km for a C-band unrepeated WDM transmission.

To increase system capacity and satisfy the constantly growing demand for bandwidth, the industry has been focusing on enabling transmission at data rates of 400G and beyond [8]. This is also the case for the unrepeated transmission scenario, with WDM transmission of 400G optical channels being defined as a possible target for future systems. To this aim, the option that represents the best trade-off between implementation cost and performance relies on dual-polarization (DP) 16QAM modulation at high symbol rates (64-GBd or higher [9]). The lower required optical signal-to-noise ratio (OSNR) when compared to other solutions, such as 43-GBd DP-64QAM, makes this option particularly attractive for unrepeated optical systems.

Recently, experimental demonstrations of single-carrier 400G unrepeated transmission were reported in [10]–[12]. The term single-carrier refers to the transmission of only one carrier per WDM channel slot in a WDM system in opposition to experiments that use superchannels (several subcarriers). In particular, the transmission of 24×400G 16QAM channels over a record distance of 443.1 km, which is to date the highest reach in unrepeated WDM transmission of single-carrier 400G signals, was demonstrated in [12]. However, that work put little detail on system design, which is extremely important especially in the unrepeated transmission scenario, where the system parameters must be carefully selected to ensure best performance.

This paper focuses on the design of single-carrier 400G

unrepeated optical systems based on ROPA stages with dedicated pump delivery fibers. Experimental results demonstrate an accurate system model for unrepeated transmission simulation that allows to evaluate the received bit error rate (BER) and OSNR with very low error. To prove the model effectiveness, the proposed method is applied to different scenarios.

II. EXPERIMENTAL DESIGN

This section details the experimental designs of two unrepeated transmission systems as in [11] and [12]. By using simulation calculations considering BER evaluation from the model presented in the Section III, experimental results can be compared to theoretical predictions. Those results were partially reported in [11] and [12] and together with this work provide a complete description of the laboratory transmission demonstrations.

The experimental setup is shown in Fig. 1. Two different configurations were considered, the first one [11], shown in Fig. 1(b), being based on single-stage unidirectional ROPA pumping, while the second one [12], shown in Fig. 1(c), relies on single-stage bidirectional pumping. Both configurations make use of dedicated fibers for remote pump power delivery to the ROPAs. To optimize system performance, different types of optical fiber were used inside the link. At the outputs of both the transmitter amplification stage (EDFA + distributed Raman amplification, DRA) and the transmitter ROPA, where the optical power per channel reaches its highest value, ultra low-loss, pure-silica core fiber with very-large effective area (A_{eff}) was used (VLA-PSCF, fiber ①, $A_{eff} = 150 \mu m^2$, loss = 0.157 dB/km), due to its higher resilience to nonlinear effects. In the midspan sections and final section in Fig. 1(b), PSCF with large effective area (LA-PSCF, fiber ②, $A_{eff} = 112 \mu m^2$, loss = 0.157 dB/km) was employed to increase system reach. In the final sections of the spans of Fig. 1(c), a terrestrial-grade low-loss single-mode fiber (LL-SMF, fiber ③, $A_{eff} =$

$82 \mu m^2$, loss = 0.183 dB/km) was used to enhance Raman gain. The pump delivery links were composed of LA-PSCF fiber only for the setup of Fig. 1(b), while for the setup of Fig. 1(c), the insertion of LL-SMF spans was required due to the lack of LA-PSCF.

1) *ROPA Optimization*: the project of an unrepeated optical system needs to take into account four amplification stages, as shown in Fig. 1: (i) a hybrid EDFA+DRA transmitter amplifier (HYB Tx); (ii) the transmitter ROPA (ROPA Tx); (iii) the receiver ROPA (ROPA Rx); (iv) a hybrid DRA+EDFA pre-amplification stage located at the receiver end (HYB Rx).

By considering single-pass, single-stage ROPAs with no gain flattening filters, the most important parameters that need to be optimized to improve system performance are: the EDFA and/or DRA configuration in the HYB Tx and HYB Rx, the erbium doped fiber (EDF) length of the two ROPAs, and their distances from the respective terminals. The optimization of the ROPA parameters is typically the most demanding in terms of computational effort. To address this issue, it is possible to divide the unrepeated system design into two steps, each one individually optimizing the transmitter and the receiver amplification stages. This approach was already proposed in [13] for dual-carrier 400G unrepeated transmission. However, in this previous work, the design methodology only took into account linear propagation effects, and transmission performance was evaluated based solely on the OSNR of the received signal. This may lead to a suboptimal solution that could greatly differ from the optimum one in systems with tighter requirements, such as in single-carrier 400G transmissions. For instance, in the latter case, system requirements tend already to be more strict from a simple theoretical B2B analysis based on Shannon limit, where the 400G OSNR is approximately 3 dB is higher than that of the 200G case at the FEC limit. To find the optimal operational point with high accuracy, the optimization process is further improved by incorporating BER evaluation of the received signal to the two-step approach, as detailed in Section

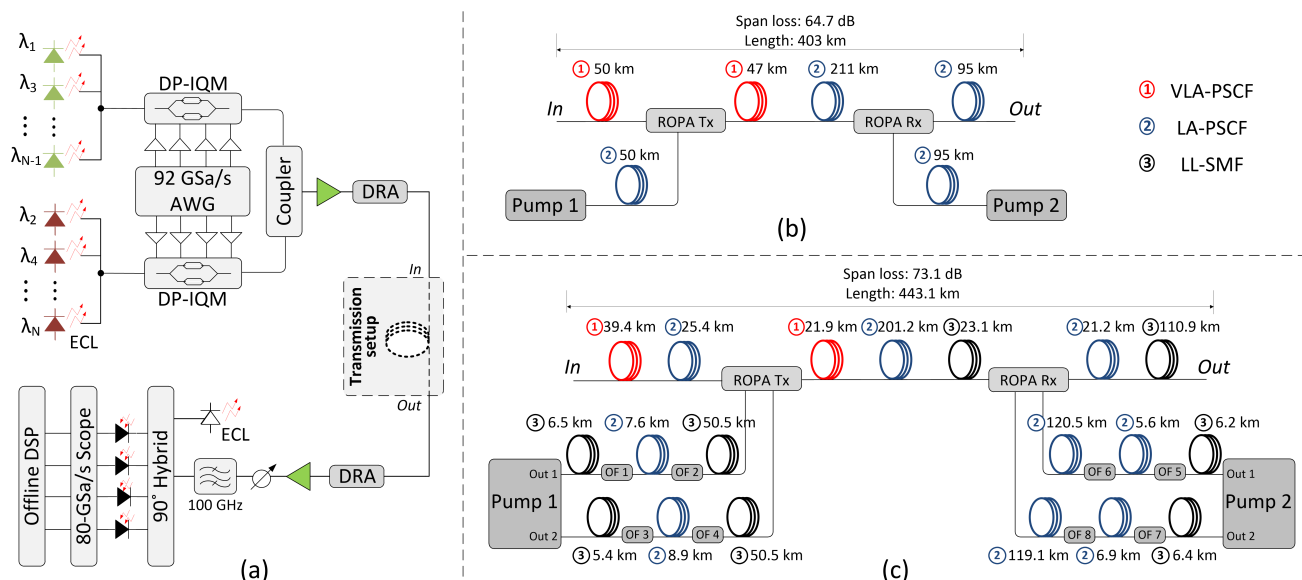


Fig. 1. Experimental arrangement: (a) transmitter and receiver stages; (b) 403-km hybrid transmission link with single-stage dedicated ROPA pumping; (c) 443.1-km hybrid link with dual-stage pumping. The transmission setups (b) and (c) depict the dashed box in (a).

IV.

When determining the ROPA Tx parameters, the saturation optical output power should be as high as possible to maximize the power level of the transmitted signal at the ROPA Tx output. However, a transmission power threshold must be defined to avoid nonlinear effects. The nonlinear threshold (NL_{th}) is directly related to the characteristics of the transmitted signal, such as pulse format, symbol rate, and channel spacing. Fig. 2 shows the influence of the NL_{th} parameter in the unrepeated system using the model of Section III. By considering both the 16 and 24×400G setups depicted in Fig. 1(b) and (c) with a total launched signal power of 15 and 17 dBm, respectively, the signal power profiles along the transmission link for different ROPA Tx configurations are presented in Fig. 2(a) and (b), each corresponding to a different value of NL_{th} .

The optimized curves in Fig. 2 correspond to distinct groups of ROPA Tx parameters (EDF length and distance from Tx terminal), highlighting the strong dependence of the ROPA Tx parameters in relation to the NL_{th} . The NL_{th} value was set at 8 dBm per channel for both transmission scenarios, as discussed later in Section III.

The ROPA Rx parameters were defined by minimizing the noise figure of the equivalent amplifier consisting of the ROPA Rx and HYB Rx [13]. For each value of system reach, the optimal ROPA Rx configuration (distance from Rx terminal and EDF length) was found, as shown between parentheses in Fig. 3 for the two transmission scenarios of Fig. 1. Depending on the OSNR requirements of the 400G optical signal, determined by computer simulations as the

minimum OSNR value to ensure worst-channel BER below the forward error correction (FEC) limit, the maximum reaches were defined without and with channel power pre-equalization at the transmitter to equalize the received OSNR per channel (see Section IV).

2) *Pump delivery link*: for unrepeated transmission based on the use of dedicated fibers to deliver the optical pump power to the ROPAs, it is possible to avoid pump-signal interference that arises when the same fiber is used for both signal transmission and remote pumping. Most relevant impairments are pump-to-signal relative intensity noise (RIN), double Rayleigh backscattering (DRB) and multipath interference (MPI). A dedicated pump delivery link also prevents signal-induced pump power depletion effects due to stimulated Raman scattering (SRS), allowing to extend system reach. The main drawback of this approach compared to a shared pump-signal transmission scheme is the higher cost. Nevertheless, for some critical applications system efficiency can be more important than system cost. For instance, pump power delivery using dedicated fibers represents a very attractive solution for challenging terrestrial applications in the energy, gas, and oil industries.

From a system design point of view, the more pump power it is launched into the transmission link, the further from the terminals it is possible to locate the ROPAs, which results in longer transmission reaches. However, even when a dedicated pump delivery link is used, the high pump power level that is launched into the fiber can give rise to SRS. In this case, the fiber acts as a gain medium for the Stokes-shifted wavelength region associated with the pump wavelength,

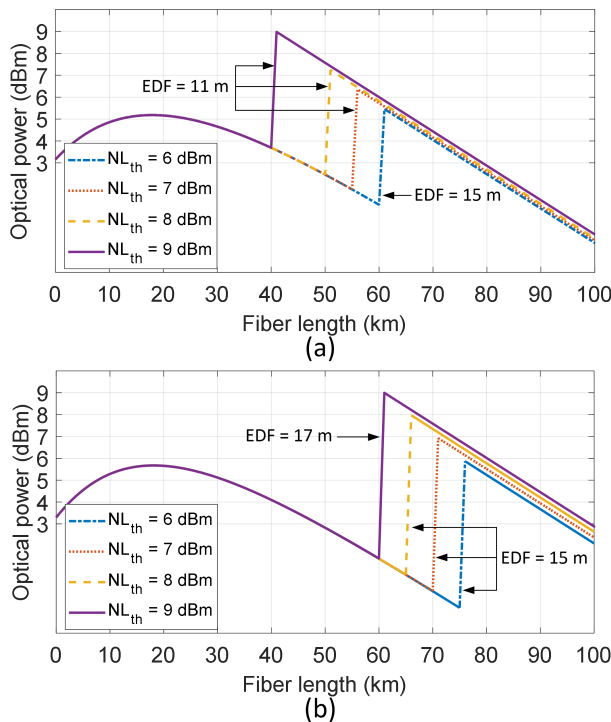


Fig. 2. Simulated channel power profile along the transmission link and corresponding ROPA Tx EDF lengths considering different NL_{th} , for (a) 16×400G×403-km and (b) 24×400G×443.1-km transmissions.

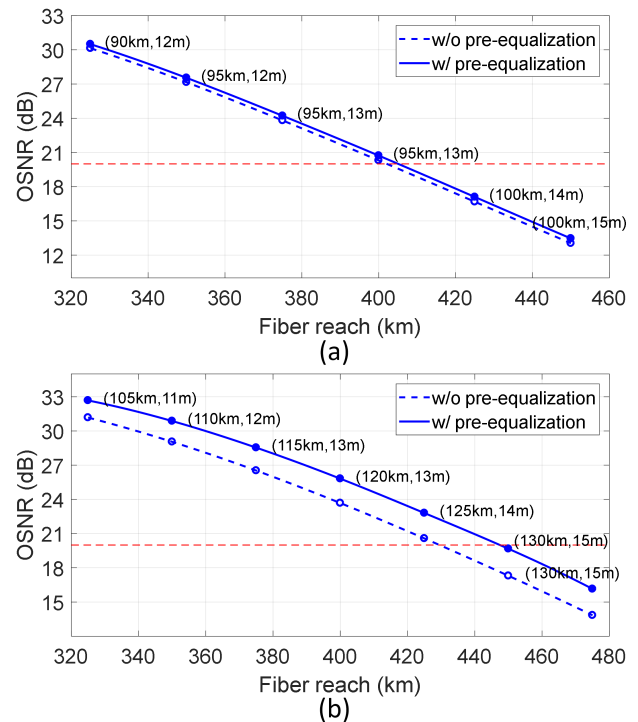


Fig. 3. Simulated received OSNR vs. system reach, with (solid line) and without (dashed line) channel power pre-equalization for (a) 16×400G×403-km and (b) 24×400G×443.1-km transmissions.

causing pump power depletion. Besides that, the SRS-induced gain, in conjunction with discrete and distributed reflections that may occur along the pump delivery link, can originate a cavity-like behavior inside the fiber, a phenomenon commonly known as lasing. When the lasing threshold is reached (i.e. the total SRS gain in the fiber cavity overcomes the losses), stochastic peaks at the Stokes wavelengths (which fall within the signal window) can be observed. Unfortunately, their total power can be considerably higher than that of the background amplified spontaneous emission (ASE) noise [14], leading to amplification instabilities in the ROPA and degradation of the transmitted signal.

To avoid lasing inside the pump delivery link, the launched pump power into the fiber must be kept below the lasing threshold. To increase this threshold and allow higher pump power levels to be launched, some alternatives can be considered. One possibility is to use low-loss, large effective area fibers with low SRS gain and low Rayleigh backscattering coefficients. Also, it is important to carefully manage all fiber splices in the delivery link to minimize discrete reflections. Finally, it is recommended the use optical filters (OFs) centered at the ROPA pump wavelength region (around 1480 nm) in the delivery link to suppress the SRS-induced Stokes gain. In Fig. 1(c) (bidirectional ROPA pumping), two OFs were used in each pump delivery link. The filters are usually placed at the beginning of the link, since most of the Raman gain occurs within the effective length of the optical fiber.

The lasing threshold was determined empirically for each delivery link. The output power of the pump laser was increased, and the optical spectrum corresponding to each launched power level was observed at the link end. Fig. 4 and 5 show the optical spectra at the outputs of the transmitter and receiver delivery links, respectively. When the launched power level reaches the lasing threshold, random spikes and dips appear over the background ASE close to the signal wavelength region. Based on experimental measurements, the optimal launched pump power per delivery link was 3.2 W for the ROPA Tx and 3.5 W for the ROPA Rx. The higher lasing threshold level for the receiver delivery links is caused by the fact that they are prevalently consisted of LA-PSCF, while LL-SMF was predominant for the transmitter delivery links.

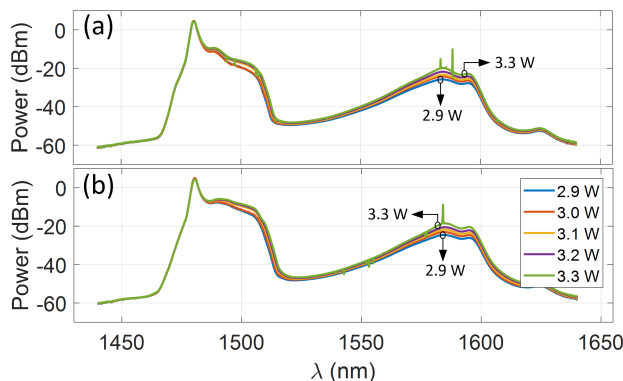


Fig. 4. Optical spectra at the output of ROPA Tx delivery link for the unit pump source out (a) 1 and out (b) 2. Lasing induced fluctuations can be observed at 3.3-W pump power.

3) *Transmitter, coherent receiver and DSP*: this section details the designed architectures of the transmitter and receiver shown in Fig. 1(a) used later in the experimental demonstrations. To generate the WDM signals at the transmitter side, two arrays of external cavity lasers (ECLs) (100-kHz linewidth) were coupled to and independently modulated by a pair of 29-GHz LiNbO₃ dual-polarization in-phase quadrature modulators (DP-IQM) to generate the 400G single-carrier DP-16QAM channels. Each DP-IQM is fed by four independent electrical signals previously amplified by linear drivers (35-GHz bandwidth, SNR = 21 dB at 32 Gb/s PRBS31 input) and generated by a 8-bits resolution arbitrary waveform generator (AWG) with 92-GSa/s sampling rate, 32-GHz bandwidth and 4 ps_{pp} of total jitter. Raised-cosine (RC) pulse shaping was employed with variable roll-off factor, as discussed in the following. In the experimental demonstration associated with the setup in Fig. 1(b), 16 channels were transmitted, with a symbol rate of 66 Gbaud, 0.1 roll-off factor and 75 GHz channel spacing, yielding a SE of 5.33 b/s/Hz. In the experimental investigation associated with the setup in Fig. 1(c), 24 channels were transmitted. Two different options for single-carrier 400G channels generation were considered: (i) 66-Gbaud symbol rate, 0.1 roll-off factor, and 75-GHz channel spacing and (ii) 64-Gbaud symbol rate, 0.05 roll-off factor, and 68.75-GHz channel spacing, the latter configuration resulting in an improved SE of 5.81 b/s/Hz. In all cases, the frequency of the first channel was set at 191.65 THz. Before being transmitted by the optical link, the WDM channels were amplified by a hybrid amplifier stage (HYB Tx) consisting of a booster EDFA followed by a DRA using two forward Raman pumps at 1452 and 1457 nm, with 700-mW total launched pump power. At the receiver side, each 400G channel is selected by a tunable bandpass filter and detected using a polarization-diversity coherent receiver, consisting of a 90° optical hybrid connected to a local oscillator ECL with 100-kHz linewidth, followed by four 40-GHz balanced photodetectors (BPD). The electrical signals at the BPD outputs were sampled at 80 GSa/s by a 4-channel 8-bits resolution real-time oscilloscope (35-GHz bandwidth, 2 ps_{rms} of total jitter) before off-line DSP.

The DSP included resampling (2 samples per symbol), orthonormalization, static chromatic dispersion compensation,

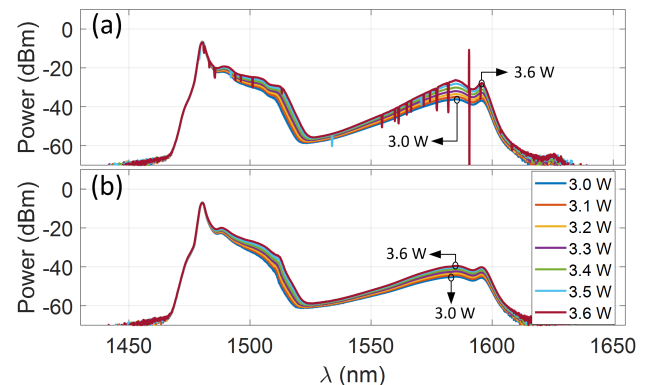


Fig. 5. Optical spectra at the output of ROPA Rx delivery link for the unit pump source out (a) 1 and out (b) 2. Instabilities begin to appear when the pump power level approaches 3.6 W.

36-tap decision-directed least mean square (DD-LMS) based dynamic equalization and carrier recovery. For FEC, a soft-decision left-terminated, spatially-coupled low-density parity-check code (SC-LDPC) [15], with syndrome former memory $\mu = 2$, was used. From the experimental transmission data, the counted post-FEC BER was calculated using the same method as in [16] for all individual channels. The aim was to verify if the SC-LDPC could maintain error-free operation with a specific symbol-rate overhead (OH) soft-decision FEC (31%-OH SD-FEC for 66 GBd in 75-GHz grid and 27%-OH SD-FEC for 64 GBd in 68.75-GHz grid).

III. SYSTEM MODELING

In this section, the system model that was developed to simulate the transmission of WDM optical signals over an unrepeaters fiber link is presented. The system parameters were defined according to the architecture described in Section II and detailed in Fig. 1. The transmitter in the numerical simulator was configured to generate a 2^8 random bit sequence that was used to create the four components of a PDM-16QAM signal at the output of a digital-to-analog converter (DAC). The modulation is accomplished by sending these components to the I and Q ports of the X and Y stages of DP-IQM. At the transmitter output, the optical signal is directed to a first amplification stage, consisting of either a single booster EDFA or an EDFA in conjunction with a DRA stage. Two ROPA stages are included in the transmission fiber link, one pumped from the transmitter and the other one from the receiver ends. At the receiver side, an EDFA preamplifier or a DRA+EDFA stage amplifies the incoming signal before sending it to the coherent receiver. The receiver consists of a local oscillator (LO) laser followed by a 90° optical hybrid, whose outputs are connected to four BPDs. The electrical signals from the BPDs are then sampled by analog-to-digital converters (ADCs) before off-line processing. The laser phase noise, as well as jitter, effective number of bits, quantization and clipping error of the DAC and ADC, are directly or indirectly included in the simulation model. Also, the bandwidth limitation of all the system components is considered during simulations. All the component parameters were defined based on experimental measurements or vendor specifications (see Section II).

The proposed simulation model for unrepeaters transmission systems considered the major penalties of optical and electronic devices and components used in the experimental arrangement at both the transmitter and receiver. Furthermore, the same digital signal processing algorithms were applied in both off-line experimental and theoretical analyzes, allowing for a fair comparison between BER values produced from both instances. As a result, equivalent system performances were achieved in back-to-back (B2B) configurations, as shown in Fig. 6. As it can be seen, simulated and experimental results present reasonable agreements for both evaluated symbol rates. More importantly, in the region around the FEC limit, which has most significance in contemporary transmission systems, the OSNR difference between the results is lower than 0.3 dB (inset).

The erbium doped fiber based amplification stages (booster, ROPAs and pre-amplifier) were implemented using finite-

difference time-domain (FDTD) method and relied on a two-level model of EDFAs proposed in [17], describing core- or cladding-pumped amplifiers under significant ASE level and taking into account different emission and absorption cross sections for pump and signal fields.

The signal propagation over the fiber link was numerically evaluated using the coupled nonlinear Schrödinger equation (NLSE) together with Manakov system approximation to average the modal birefringence effect [18]. The NLSE was solved using the split-step Fourier (SSF) method. Intrachannel and interchannel nonlinear effects, such as self-phase modulation (SPM) and cross-phase modulation (XPM), were included in the simulation model, where the fiber transmission step-size was properly chosen (1 km) to provide accuracy to the propagation solution. The SRS contribution was also included and separately modeled as a set of coupled power equations [19] solved via the 4th order Runge-Kutta method. Since the same step size of 1 km was used in all evaluated numerical models, it was possible to use the noise and signal power profiles over the fiber link calculated from the SRS coupled equations in the SSF based model. More specifically, after each SSF step, the correspondent DRA signal gain could be considered in order to properly evaluate the effect of SPM and XPM. The ASE noise from the two DRA stages was modeled as a Gaussian noise and added to the signal at the end of the transmission link. This approximation is justified by the fact that, in the scenario under consideration, the signal power is significantly higher than that of the noise, thus ensuring a sufficiently high OSNR during signal transmission. For this reason, nonlinear interactions between noise and signal can be ignored.

In the proposed model, only SPM, XPM and SRS nonlinearities were taken into account. This is because, as illustrated in Fig. 1 and explained in the previous section, the analysis was targeted to unrepeaters systems with dedicated delivery fibers for ROPA pumping and in which DRA is limited to first-order. Also, the use of low-loss, large effective area fibers with high dispersion ($D \approx 20$ ps/(km·nm) at 1550 nm) is considered. This high-dispersion coefficient fiber considerably reduces inter-channel four-wave mixing induced penalties, as phase matching conditions are no longer satisfied in the fiber. In the same way, DRB and, consequently, MPI effects could be neglected since low-loss and large effective area fibers,

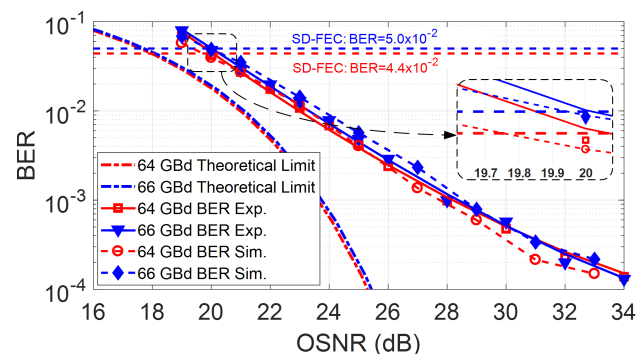


Fig. 6. BER vs. OSNR curves in B2B for 64 (red curves) and 66 GBd (blue curves) symbol rates, considering experimental (Exp.) and simulation (Sim.) of one-channel transmission.

as well as dedicated ROPA pump delivery links, were used. Besides, [20] shows that under first order DRA amplification the dominant effect in unrepeated systems is the SPM, which can present five times as large contribution as that of DRB. Moreover, since the Raman pump laser sources have a low RIN levels (≈ -140 dB/Hz), signal degradation induced by first-order Raman amplification is negligible, as demonstrated in [21].

The system model adopted in this work does not take into account cross-polarization effects, for which the main contributions are the cross-polarization modulation (XPolM) and the pump-to-signal cross-polarization scattering (XPolS). Regarding cross-polarization modulation (XPolM), although such effect might induce phase noise and polarization crosstalk between the two orthogonal polarization components of a PDM signal, for amplitude-varying modulation formats and high symbol rate signals, as in the cases under study (400G 16-QAM), the impact of XPM is much larger than that of XPolM [22]. To corroborate that, [21] demonstrates that the phase noise induced by XPolM is nine times as small as that of the XPM, thus allowing XPolM contribution to be neglected during simulations. On the other hand, in systems using DRA, the XPolS can be induced by the state of polarization and power fluctuations of the pump laser [19], [20]. Once DRA pump depolarization is guaranteed in both simulation model and experimental demonstrations, the XPolS can be reduced to a RIN induced impairment [20]. Since RIN is not an issue in the proposed model, XPolS contribution can also be discarded. Due to the short transmission reach of the proposed single-carrier 400G unrepeated optical system (a few hundred kilometers), the effect of polarization mode dispersion (PMD) can also be disregarded in the model. Since cross polarization effects are discarded and PMD is negligible, each one of the PDM axes used during the experiment are considered as independent from each other during simulations. In this way, each transmitted polarization axis can be modeled as if it were in a single polarization transmission case.

By considering the nonlinear effects previously addressed, an important parameter that was defined in Section II was the NL_{th} . Its value was originally calculated based on empirical observations, after considering NL_{th} results (8 dBm) reported in [2] and [3] for systems based on large area fibers. The NL_{th} was chosen to be 8 dBm after checking system performance for different NL_{th} values around 8 dBm. Later, the simulation algorithm was used to confirm the experimental findings. Since the unrepeated system under analysis is based on ROPAs, as pointed out in Section II-1, there is a high dependence between ROPA Tx design and the NL_{th} . Thus, to evaluate system performance for different NL_{th} settings, firstly, ROPA Tx must be optimized (EDF length and distance to Tx) for the corresponding NL_{th} value. After that, the resulting BER is calculated for all transmitted channels. The best BER performance defines the optimum NL_{th} . To illustrate that, Fig. 7 shows BER calculated results for Channel 9 (central) as a function of NL_{th} , for the simulation of the experimental arrangement of Fig. 1(b). ROPA Tx EDF length and distance to Tx are presented inside the parenthesis. The dashed line represents the interpolation of the calculated points. Similar

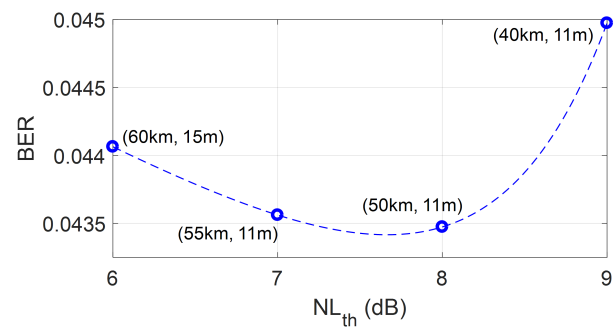


Fig. 7. Calculated BER for Channel 9 versus NL_{th} for the simulation of the experimental arrangement of Fig. 1(b), having ROPA Tx EDF length and distance to Tx as parameters; the dashed line represents the interpolation of the calculated points.

behavior is expected for the experimental arrangement of Fig. 1(c), since the same type of large area fiber (VLA-PSCF) was used in the parts of the transmission link where the optical power is high (higher nonlinearity). For the values of NL_{th} investigated (circles in Fig. 7), it is possible to observe that the best system performance is obtained when NL_{th} is close to 8 dBm.

IV. UNREPEATED TRANSMISSION PERFORMANCE

Once the parameters of the two ROPAs (position and EDF length) had been defined according to the methodology described in Sections II and III, the next step in the unrepeated system design consisted of defining a suitable range of launched signal power levels to identify the best transmission scenario. For each level of total launched power, channel power pre-equalization at the transmitter was performed to attain similar OSNR values for all channels at the receiver. In the following, simulation and experimental results are compared for both transmission scenarios. Computer simulations were used to find the best launched power levels for each case under study and a good agreement with the experimental results was observed. Fig. 8 shows the experimental B2B curves for 64 (red curves) and 66 GBaud (blue curves) symbol rates. The required OSNR to achieve a BER below the pre-FEC limit was ≈ 20 dB (inset) for both cases. Negligible penalty was observed between the one-channel (OC) and WDM (measured at central channel) configurations.

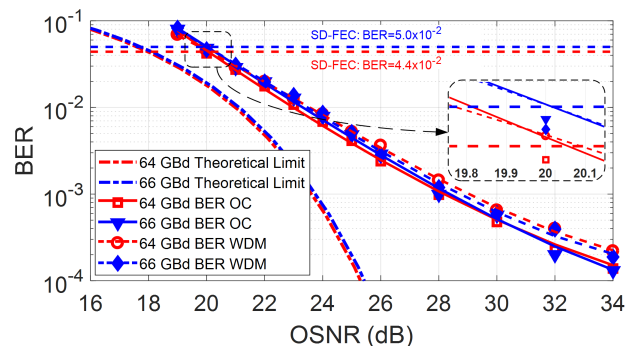


Fig. 8. BER vs. OSNR curves in B2B for 64 (red curves) and 66 GBaud (blue curves) symbol rates, considering one-channel (OC) and WDM configurations.

1) *16x400G setup*: the first evaluated transmission scenario was the one illustrated by the setup in Fig. 1(b). The experimental demonstration of the $16 \times 400\text{G}$ single-carrier transmission over a hybrid link of 403 km has already been reported in [11]. Here, a more detailed analysis of the system design phase is provided, which led to the definition of the optimal set of parameters used in the experimental demonstration. Fig. 9(a) shows the BER results for five selected channels (2, 5, 9, 12, and 15), evaluated by simulation, as a function of the total launched power. Channel power pre-equalization was performed for all cases to ensure a flat received OSNR.

It is important to point out the influence of power pre-equalization on the transmission performance. Since a typical EDFA delivers higher gain at higher frequencies, channels at those frequencies tend to be more susceptible to ASE noise contributions and thus are more penalized due to lower received OSNR. As a result, it is necessary to increase the launched power for those channels to guarantee the required OSNR specified by the B2B performance measurements. However, the required increase in the power level of those channels induces nonlinear transmission impairments that severely degrade system performance. This can be observed in the simulated BER curves of the five channels under analysis in Fig. 9(a). For a given value of the total launched signal power, as the channels index (frequency) increases, the corresponding BER value becomes higher. Considering a pre-FEC BER limit of 5.0×10^{-2} , it is possible to identify a 3-dB range in the launched power from 15 dBm to 18 dBm where post-FEC error-free operation can be achieved for all channels. Fig. 9(d) shows the simulated and experimental power and OSNR curves for the 16 channels with 15 dBm (blue curves) and 18 dBm (red curves) total launched power. A very good agreement between simulated and experimental results was achieved. The same levels of launched power were used to evaluate system performance both by simulation and experiment, as shown in Fig. 10(a). The simulation

and experimental curves for 15-dBm total launched power show good agreement, whereas for 18 dBm the measured even channels exhibited a slightly worse performance with respect to the simulation results. This performance difference is attributed to imperfections in the bias control circuitry of the even channel optical modulator.

2) *24x400G setup*: the second transmission setup used to evaluate the proposed unrepeaters system design method is shown in Fig. 1(c). In this case, a 24-channel transmission over 443.1 km was demonstrated, with some of the experimental results already reported in [12]. Two values of symbol rate were considered in this investigation, 64 and 66 GBaud, as already mentioned in Section II-3. The total launched signal power was varied from 10 to 21 dBm as shown in Fig. 9(b) and (c), for 64 and 66 GBaud, respectively. Six channels (2, 6, 11, 14, 19, and 23) were selected to summarize all transmitted channel performances. Moreover, channel power pre-equalization at the transmitter was again taken into account to ensure a flat received OSNR profile. For 64 GBaud, launched power levels between 14 and 17 dBm, a 3-dB range, yielded BER values below the 4.4×10^{-2} pre-FEC limit for all transmitted channels. The same levels of launched power were needed at 66 GBaud to attain BER values below the 5.0×10^{-2} pre-FEC limit for all channels.

Fig. 9(e) and (f) compare simulated and measured channel launched powers and received OSNRs for both symbol rates. Two different levels of total signal launched power after HYB Tx were considered, 17 dBm and 20 dBm. Although several channels exhibit a BER performance above the pre-FEC limit for 20-dBm launched power, this level was chosen to analyze the impact of intrachannel and interchannel nonlinearities on the transmission performance. Also in this case, channel pre-equalization led to an increase in the power level of higher frequency channels, giving rise to nonlinear impairments. This can be observed in Fig. 10(b) and (c) for 64 and 66 GBaud, respectively. The higher index channels have worse BER values,

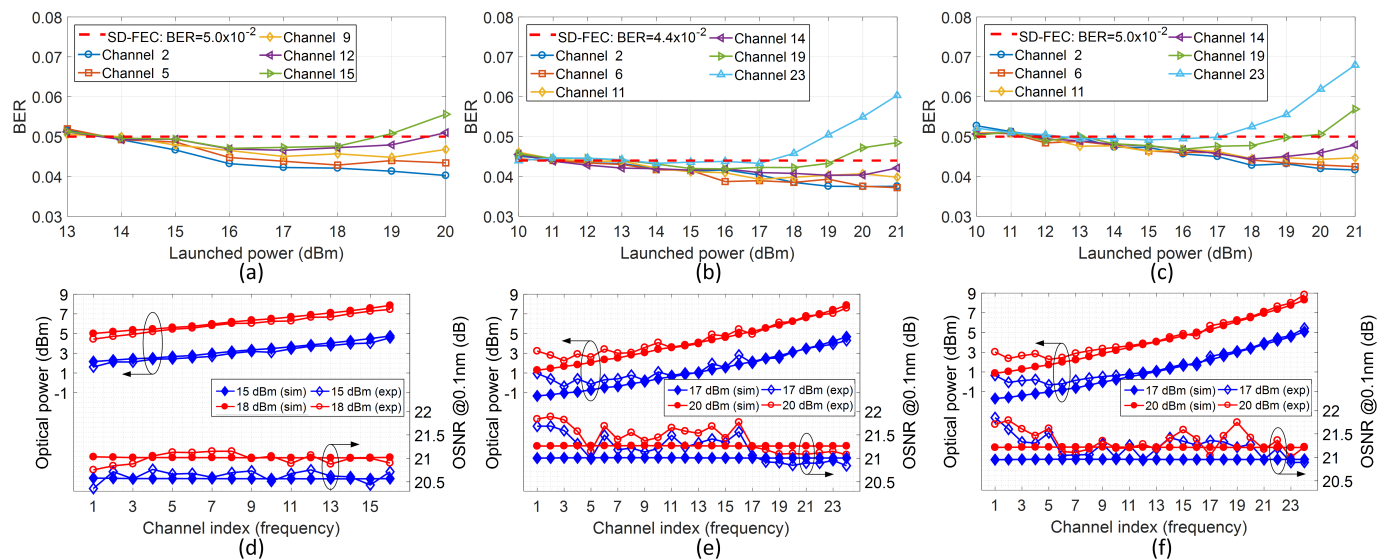


Fig. 9. Simulated BER results vs total launched power for (a) $16 \times 66\text{-GBd}$, (b) $24 \times 64\text{-GBd}$, and (c) $24 \times 66\text{-GBd}$ transmissions. For two selected total launched power levels, the respective simulated (solid marker) and experimental (empty marker) channel launched power (left axis label) and their received OSNR (right axis label) are shown in (d), (e), and (f), corresponding to the cases (a), (b), and (c), respectively.

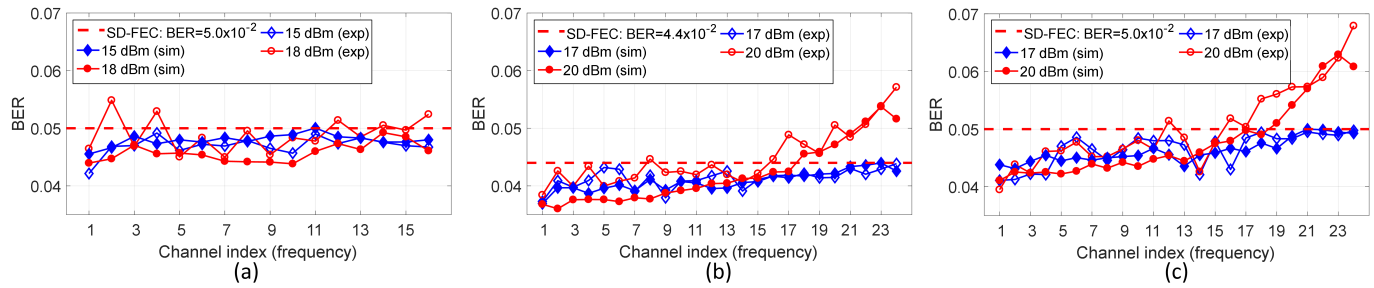


Fig. 10. Simulated (solid markers) and experimental (empty markers) BER results comparison for the considered transmission scenarios: (a) 16x66 GBd, (b) 24x64-GBd, and (c) 24x66-GBd transmissions.

despite their OSNR being practically the same as that of the lower index channels in Fig. 9(e) and (f), respectively. In [12], partial mitigation of nonlinear impairments was achieved using the digital backpropagation (DBP) algorithm to compensate for intrachannel nonlinearities. To further reduce transmission penalty, digital interchannel nonlinear compensation should also be applied, further increasing the DSP complexity. For both symbol rates and both launched signal power levels, a very good agreement was observed between simulation results and experimental measurements.

V. CONCLUSIONS

A design methodology for high-capacity (single-carrier 400G) unrepeated transmission systems was proposed and its effectiveness confirmed by experimental results. The proposed approach allows to define the optimal set of system parameters taking into account signal quality requirements together with linear and nonlinear transmission impairments. Simulation results and experimental measurements showed excellent agreement for different system scenarios making use of remote amplifier pumping over dedicated delivery fibers, which allowed to achieve the record transmission distance of 443.1 km for single-carrier 400G WDM unrepeated transmission.

ACKNOWLEDGMENT

The authors thank Stenio M. Ranzini and Alexandre Felipe for the productive discussions and the support of Brazilian Ministry of Science, Technology, Innovation and Communications (MCTIC), FUNTTEL/FINEP, CAPES, CAPES-PROEX, CNPq (306355/2014-0), FAPESP (2015/25513-6, 2016/11163-6, 311871/2016-0), and Keysight Technologies.

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